

Alkyl Halides

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Reactivity of Organic Halides in
Nucleophilic Substitution.

Organic Halides

- $R - X$ is the general formula for organic halides.

R = alkyl or aryl; X = F, Cl, Br & I

- Most of the organic halides are creatures of the laboratory.
- Only few of them have been isolated from natural products.

Alkyl Halides – Uses

Organohalogen compounds are important for several reason:

- Versatile reagents in organic synthesis.
- Can be converted into alkenes by dehydrohalogenation.
- As solvents (CH_2Cl_2 , CHCl_3)
- Insecticides, herbicides, fire retardants.
- Refrigerants and Polymers.

Classes

- Primary, Secondary, Tertiary

common organic halides

methylene chloride	CH_2Cl_2
chloroform	CHCl_3
carbon tetrachloride	CCl_4
isopropyl chloride	(2-chloropropane)
sec-butyl chloride	(2-chlorobutane)
isobutyl chloride	(1-chloro-2-methylpropane)
<i>tert</i> -butyl chloride	(2-chloro-2-methylpropane)
allyl chloride	(3-chloro-1-propene)
vinyl chloride	(chloroethene)
benzyl chloride	(chloromethylbenzene)
phenyl chloride	(chlorobenzene)

- Dihalegon derivatives (Geminal, Vicinal)
- Preparations

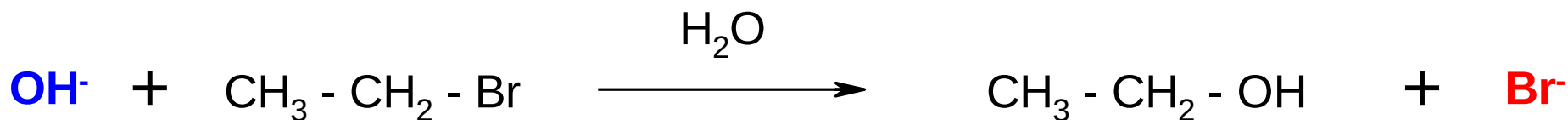
Nucleophilic Substitution

- Nucleophile is an electron rich group.
- Nucleophiles can be classified according to the kind of atom that forms a new covalent bond.
 1. Oxygen nucleophile (HO^- , CH_3O^- ...)
 2. Nitrogen nucleophiles (NH_3 , RNH_2 ...)
 3. Sulfur nucleophiles (HS^- , RS^- ...)
 4. Halogen nucleophiles (I^- ...)

Nucleophilic Substitution

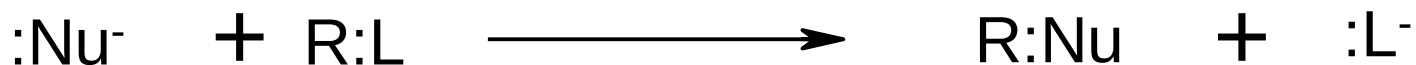
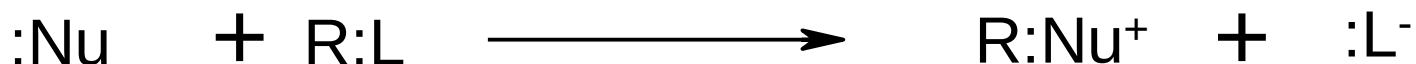
Eg:

- Ethyl bromide reacts with hydroxide ion to give ethanol.
- OH⁻ is the *nucleophile*.
- Bromide ion is called *leaving group*.
- One covalent bond is broken and one new bond is formed



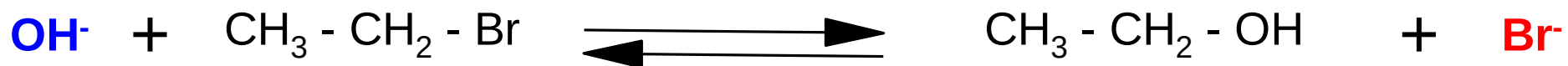
Nucleophilic Substitution

- If the nucleophile and substrate are *neutral*, the product will be *positively* charged.
- If the nucleophile is a *negative ion* and substrate is *neutral*, the product will be *neutral*.
- The unshared pair of electron in the nucleophile can be used to make new covalent bond.



Nucleophilic Substitution

- In principle, these reactions may be reversible.
- Because the leaving group also has an unshared pair of electron and that can be used to make a new covalent bond.

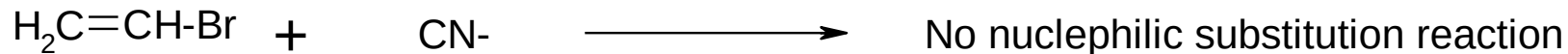
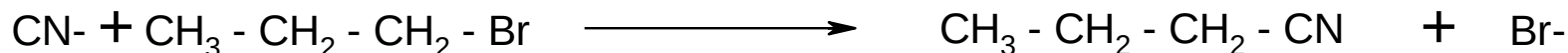


We use various method to force the reaction to go in forward direction.

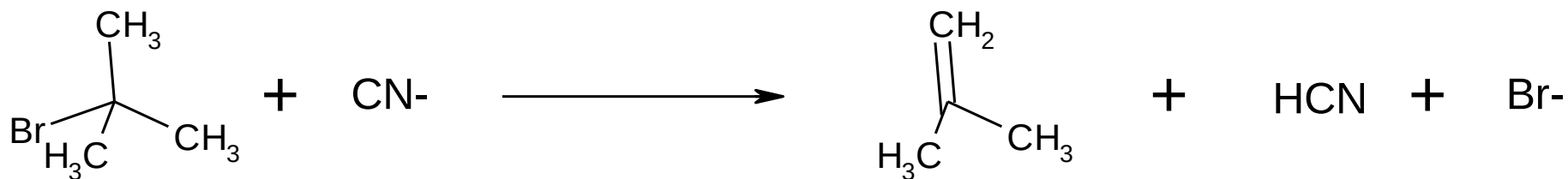
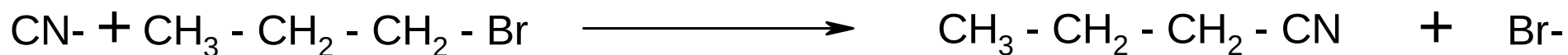
1. Stronger nucleophile compared to the leaving group.
2. Large excess of one of the reagent
3. Temperature

Nucleophilic Substitution

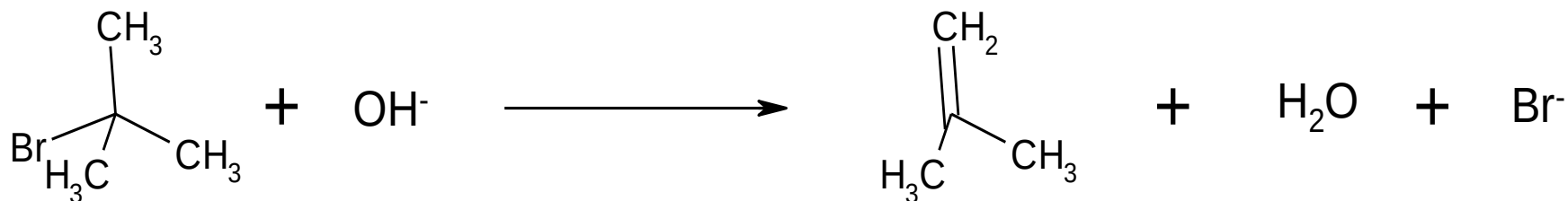
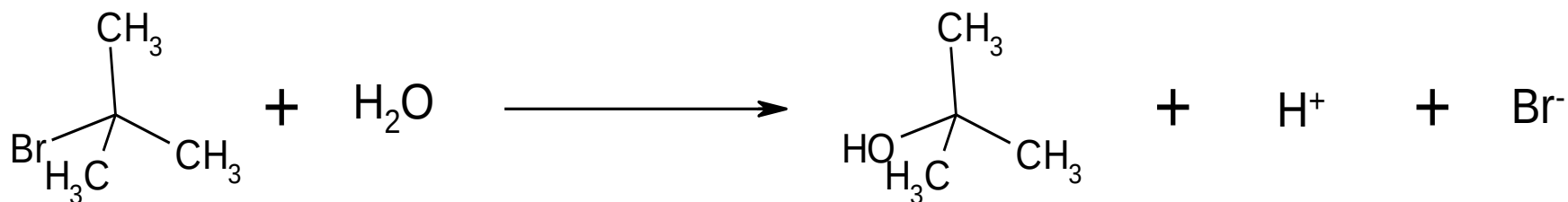
No Nucleophilic Substitution will occur on the carbon which has double bond and triple bond in which leaving group is attached



Nucleophile is same but there is difference in carbon which has leaving group.



Nucleophile is different but the carbon which has leaving group is same.



S_N Reaction Mechanisms

- “S & N” stands for Nucleophilic Substitution.

- **S_N1 Mechanism**

1 indicates that only one of the two reactants is involved in the rate determining step i.e. formation of carbocation.

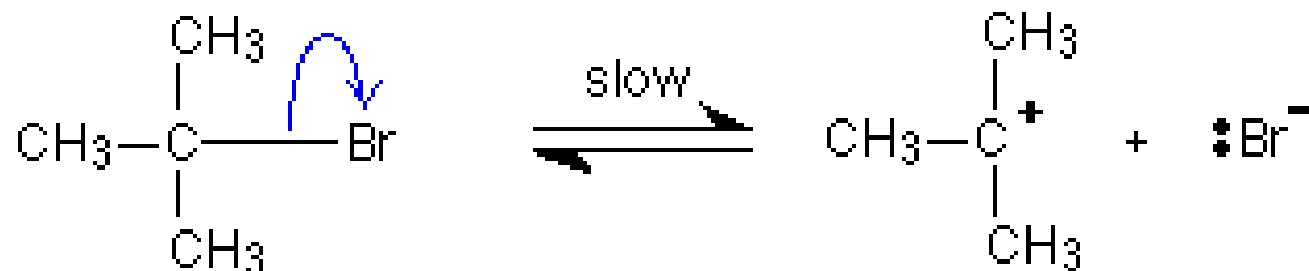
- **S_N2 Mechanism**

2 indicates the reaction is bimolecular i.e. the nucleophile and the substrate are involved in the key step

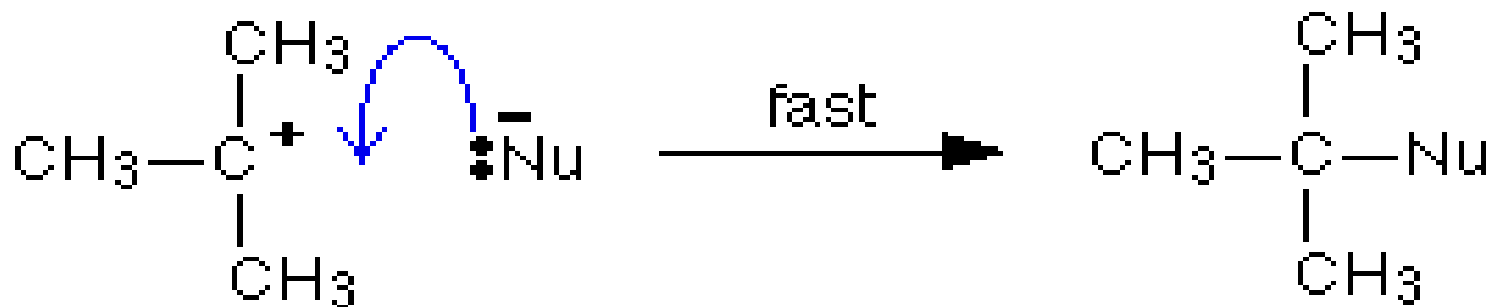
- Primary halides = S_N2
- Secondary halides = both mechanisms!
- Tertiary halides = S_N1
- Depend on: structure of a nucleophile and alkyl halide; the solvent; the reaction temperature...

S_N1 Mechanism

- It's a two step mechanism.
- Formation of the carbocation is first step and which is slow. (Ionization step)



- Second step is the fast i.e. the combination of carbocation with the nucleophile forms product.



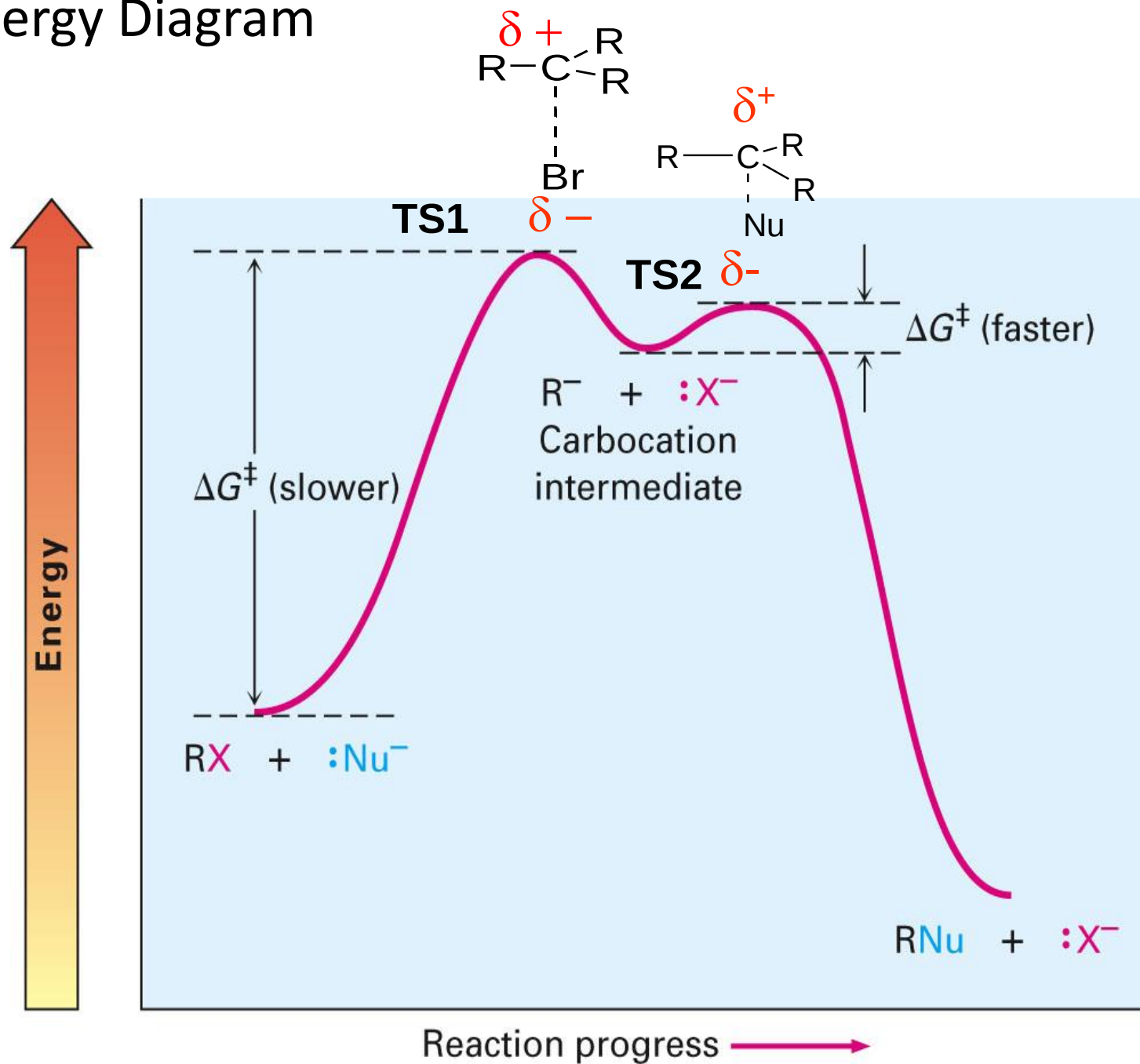
- Only one molecular species RX is involved in the heterolytic fission step which is the rate determining step. The reaction is unimolecular (SN1) and the rate R

$$R = k(R-X)$$

- Number of transition states involved in the reaction



S_N1 Energy Diagram



Stereochemistry of SN1 Reactions

Stereochemistry:

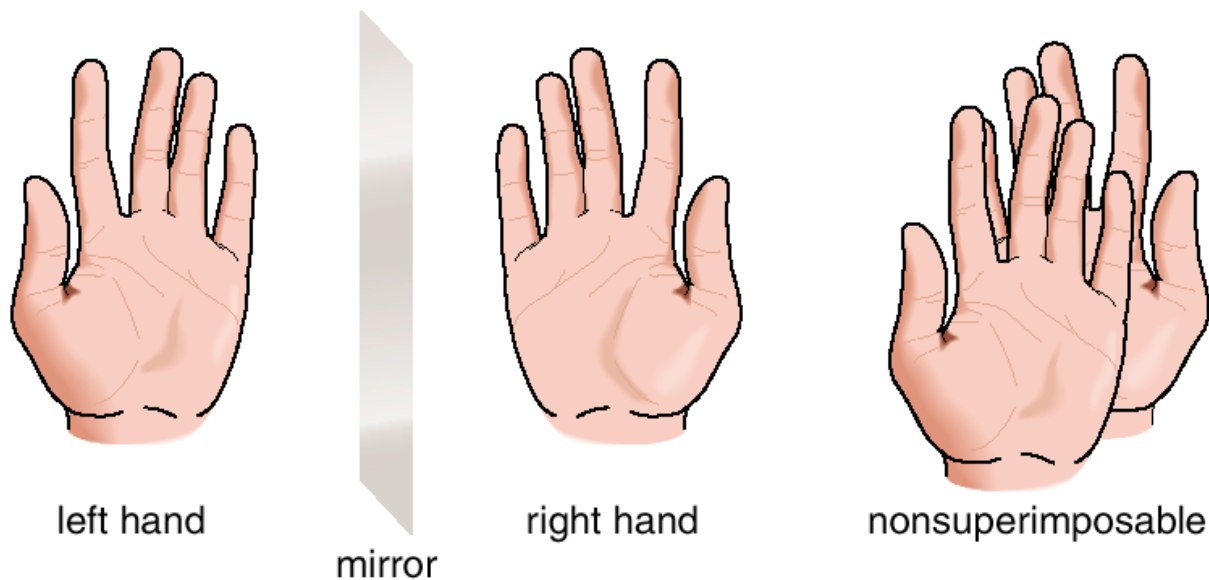
Three-dimensional arrangement of atoms (groups) in space

Stereoisomers:

Molecules with the same connectivity but different arrangement of atoms (groups) in space

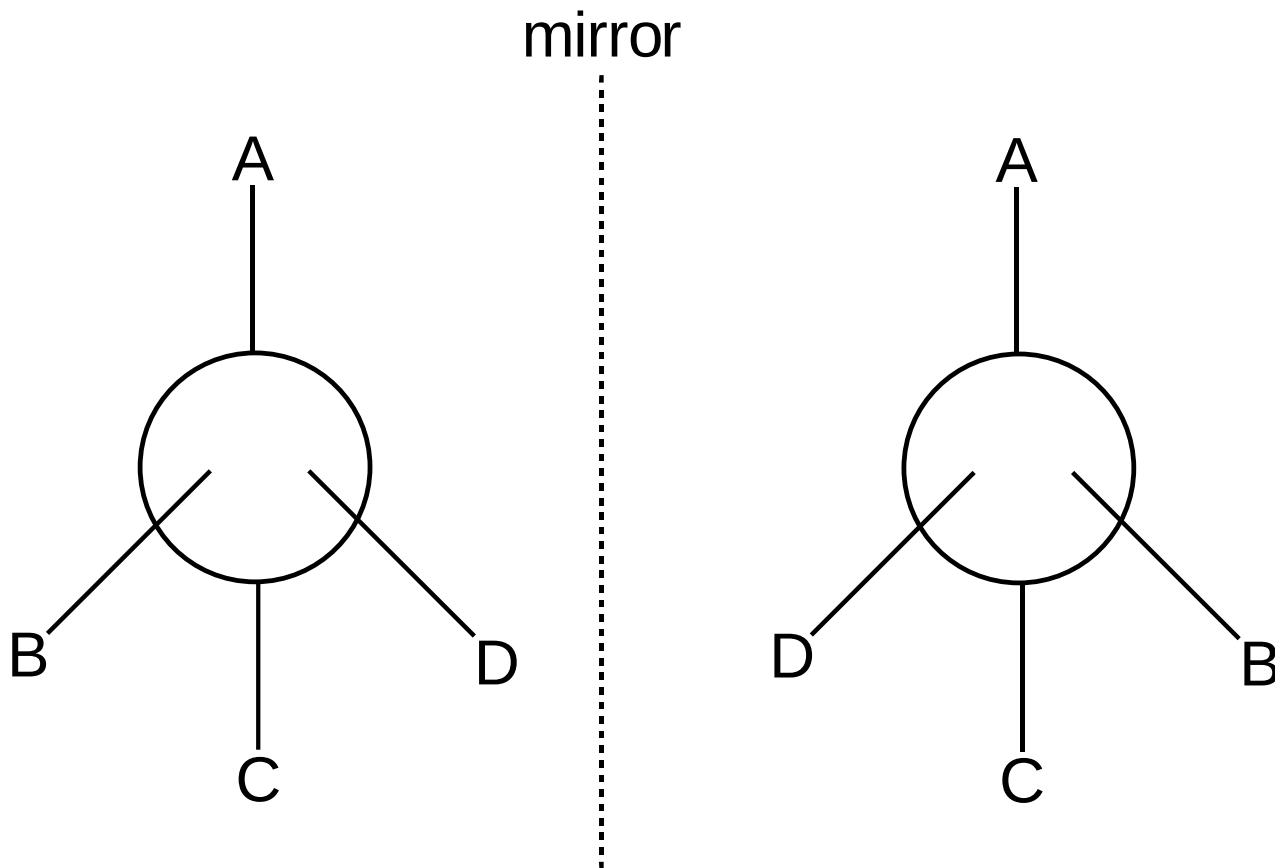
Some objects are not the same as their mirror images (technically, they have no plane of symmetry)

A right-hand glove is different than a left-hand glove. The property is commonly called “handedness”



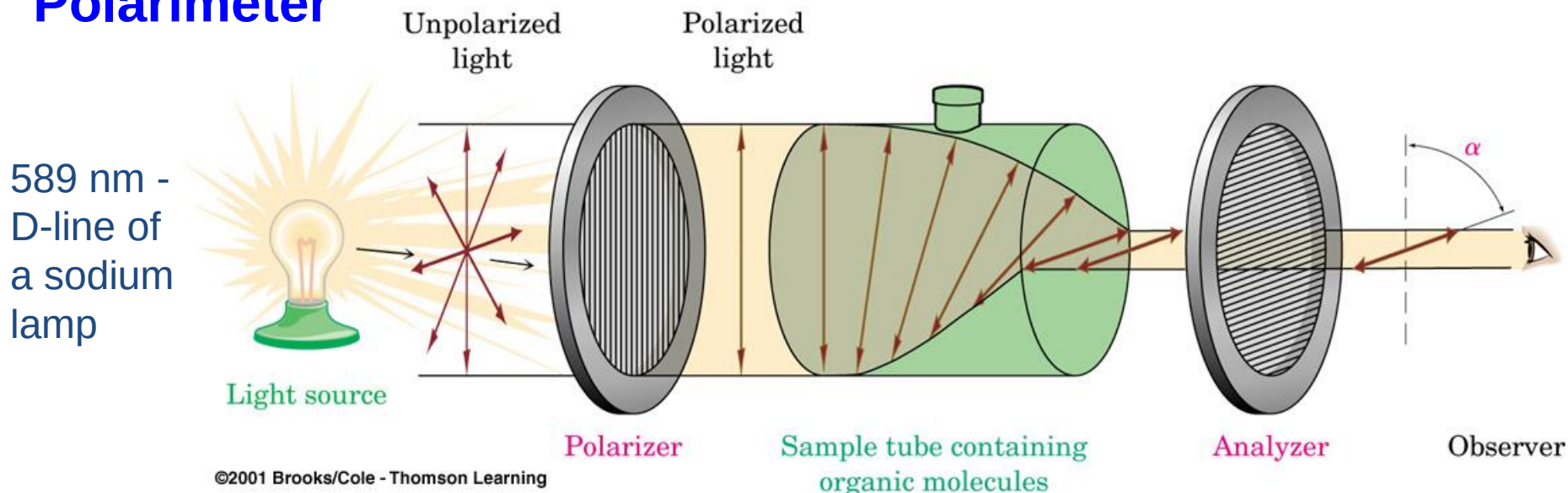
- A molecule (or object) that is *not* superimposable on its mirror image is said to be *chiral*.

The two are non-superimposable, mirror images. Such isomers are called **enantiomers**.



Optical Activity - molecules enriched in an enantiomer will rotate plane polarized light are said to be *optically active*. The optical rotation is dependent upon the substance, the concentration, the path length through the sample, and the wavelength of light.

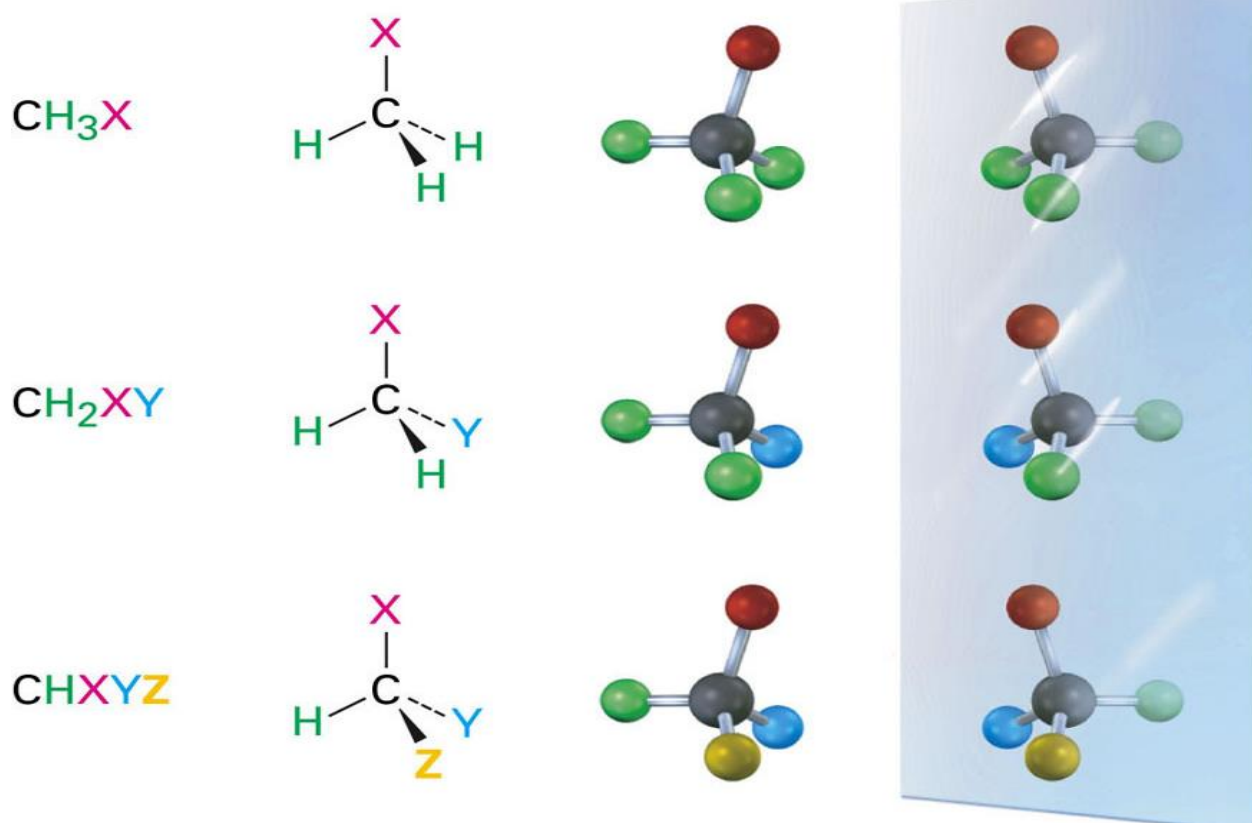
Polarimeter



Plane polarized light: light that oscillates in only one plane

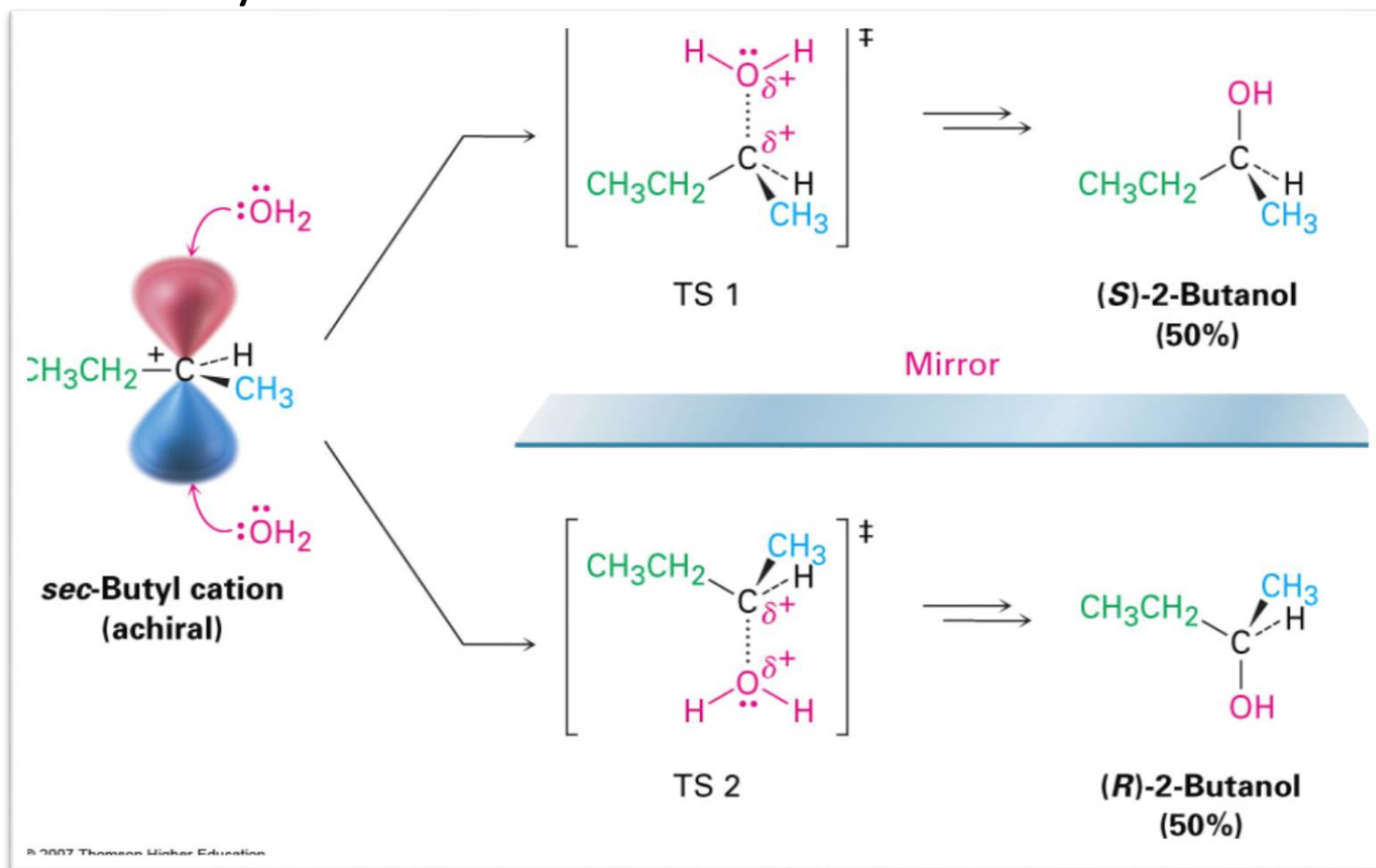
Examples of Enantiomers

- Molecules that have one carbon with 4 different substituents have a non-superimposable mirror image
- **Enantiomers** = non-superimposable mirror image stereoisomers
- Enantiomers have identical physical properties (except for one)
- Build molecular models to see this



Achiral Intermediate Gives Racemic Product

- Addition via carbocation
- Top and bottom are equally accessible
- Achiral reactant + Achiral reactant = Optically Inactive Product
- Optical Activity doesn't come from nowhere

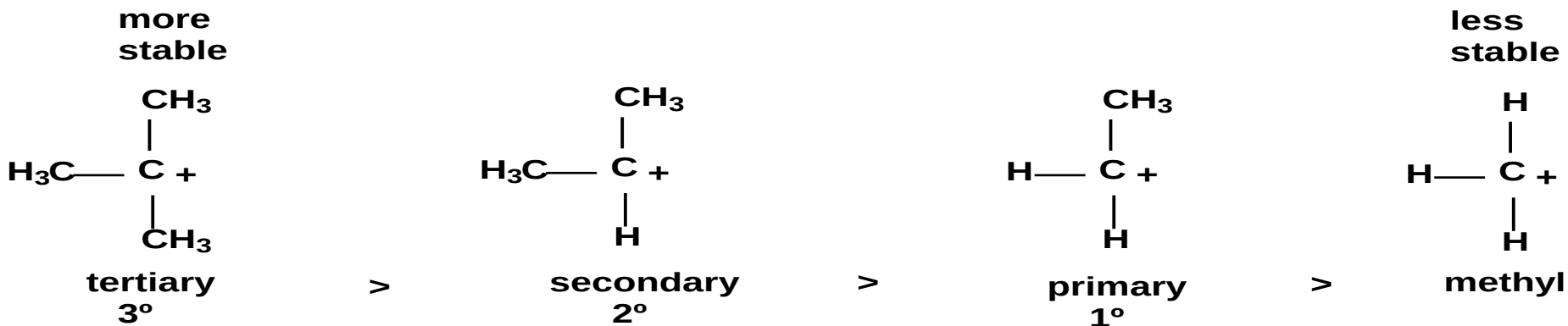


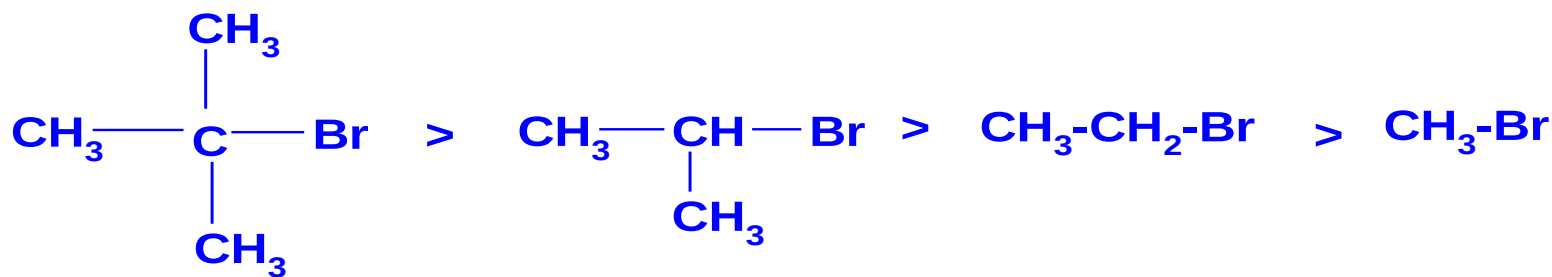
- The unimolecular reaction proceeds through an intermediate carbocation which is planar (sp^2 hybridized C atom). It is therefore expected that a chiral alkyl halide will give a racemic alcohol since the carbocation has an equal chance of being attacked by the nucleophilic reagents from either side of the plane.
- The total racemization takes place in SN_1 reaction. If the attack of the nucleophile on the two faces are purely random, it would expect to obtain equal amount of two enantiomers (racemic mixture)
- An optically pure (+) or (-) substrate give a product which is only about 50% or less optically pure. In other words, the starting material contains only one enantiomer whereas the product contains both.

Factors influencing Rate of S_N1 Reaction

- Rate of the reaction does not depend on the concentration of the nucleophile.
- It depends only on the concentration of the substrate.
- Reaction is fastest when the alkyl group of the substrate is tertiary and intermediate for secondary and slower for primary.
- Because formed carbocation is more stable in tertiary carbon compared to primary. $3^\circ > 2^\circ > 1^\circ$

increasing rate of S_N1 reactions

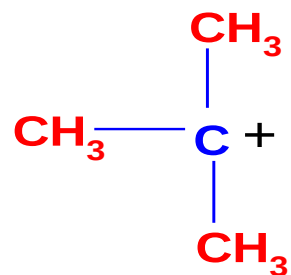




tertiary

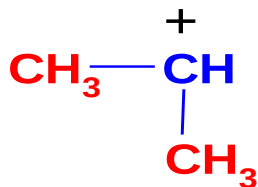
secondary

primary



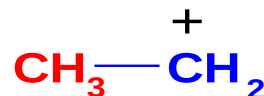
tertiary
carbocation
(very stable)

three methyl
groups



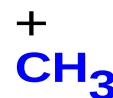
secondary
carbocation

two methyl
groups



primary
carbocation
(unstable)

one methyl
group



very unstable
carbocation

no methyl
groups

Effects of Leaving Group

An SN1 reaction speeds up with a good leaving group (leaving group is involved in the rate-determining step)

A good leaving group wants to leave so it breaks the C-Leaving Group bond faster. Once the bond breaks, the carbocation is formed and the faster the carbocation is formed, the faster the nucleophile can come in and the faster the reaction will be completed.

A good leaving group is a **weak base** because weak bases can hold the charge. They're happy to leave with both electrons and in order for the leaving group to leave, it needs to be able to accept electrons.

Strong bases, on the other hand, donate electrons which is why they can't be good leaving groups.

Excellent	• TsO^- • NH_3
Very Good	• I^- • H_2O
Good	• Br^-
Fair	• Cl^-
Poor	• F^-
Very Poor	• OH^- • NH_2^-

As you go from left to right on the periodic table, electron donating ability decreases and thus ability to be a good leaving group increases.

Halides are an example of a good leaving group whos leaving-group ability increases as you go down the column



Best

Worst

(para-toluenesulfonate = "tosylate", TsO^-).

The two reactions below is the same reaction done with two different leaving groups. One is significantly faster than the other. This is because the better leaving group leaves faster and thus the reaction can proceed faster.



Effect of Solvent

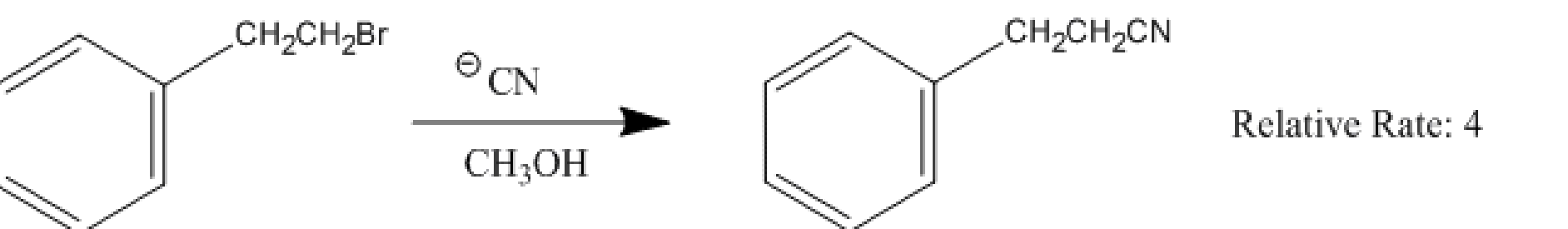
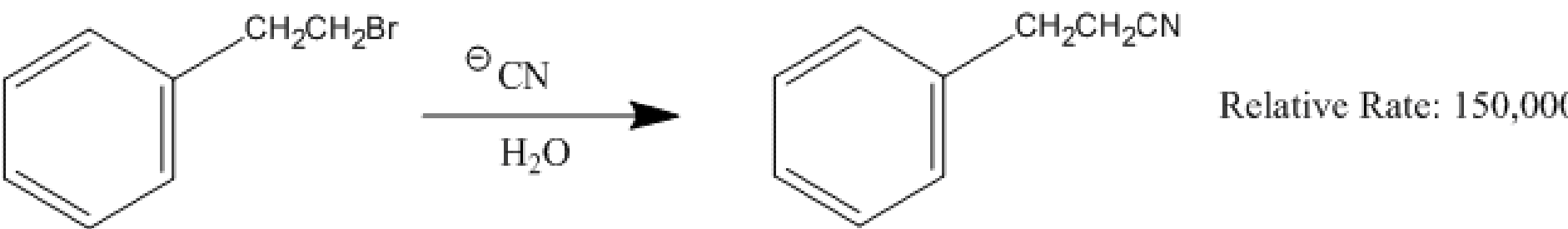
- Sometimes in SN1 reactions, the solvent acts as the nucleophile (solvolysis reaction)



- The polarity and the ability of the solvent to stabilize the intermediate carbocation is very important
- A dielectric constant below 15 is usually considered non-polar. Basically, the dielectric constant can be thought of as the solvent's ability to reduce the internal charge of the solvent.
- Higher the dielectric constant the more polar the substance and in the case of SN1 reactions, the faster the rate.

Solvent	Dielectric Constant	Relative Rate
• CH₃CO₂H • CH₃OH • H₂O	• 6 • 33 • 78	• 1 • 4 • 150,000

Below is the same reaction conducted in two different solvents and the relative rate that corresponds with it.



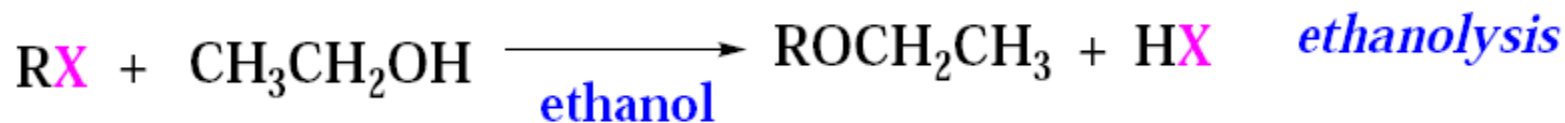
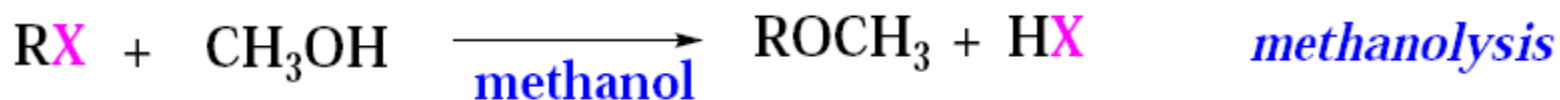
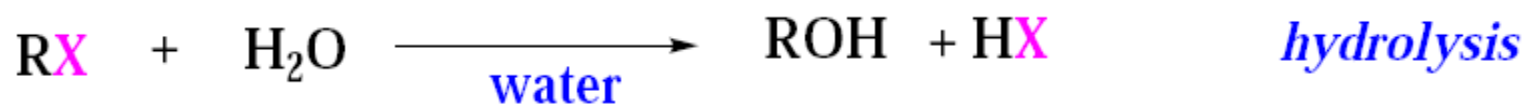
polar protic solvents are: acetic acid, isopropanol, ethanol, methanol, formic acid, water, etc

Solvolysis

A nucleophilic substitution reaction where the solvent is the nucleophile is called **solvolysis**.

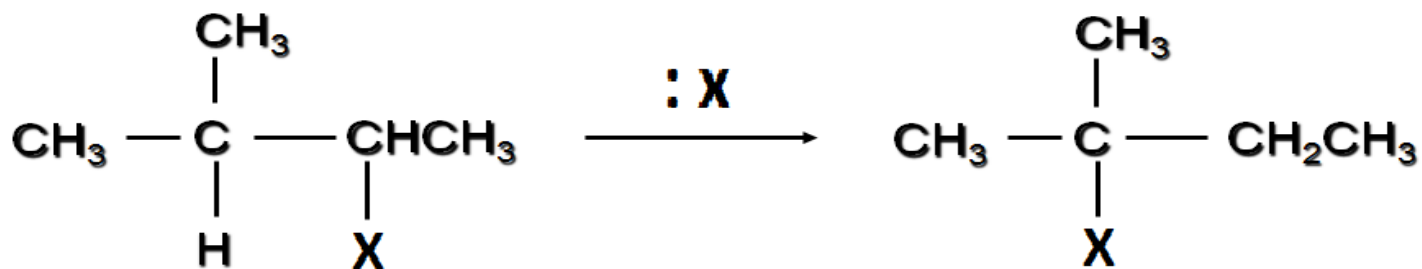
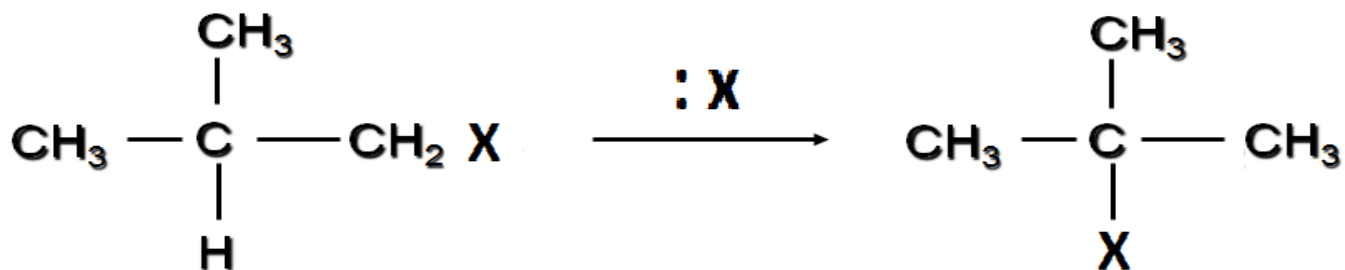
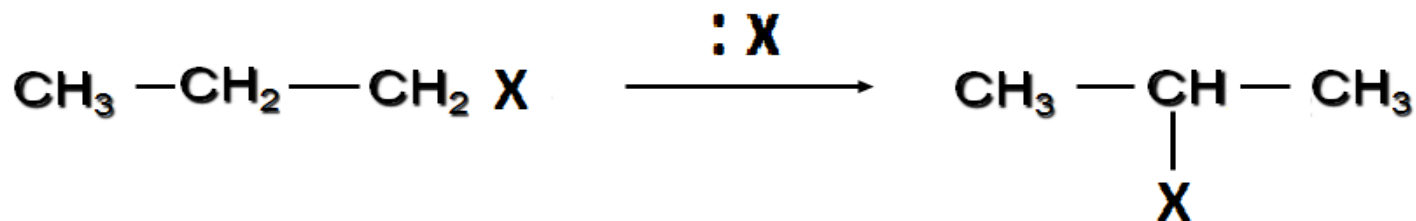
When water is the solvent, the term **hydrolysis** is used.

Methanolysis means methanol is the nucleophile/solvent.

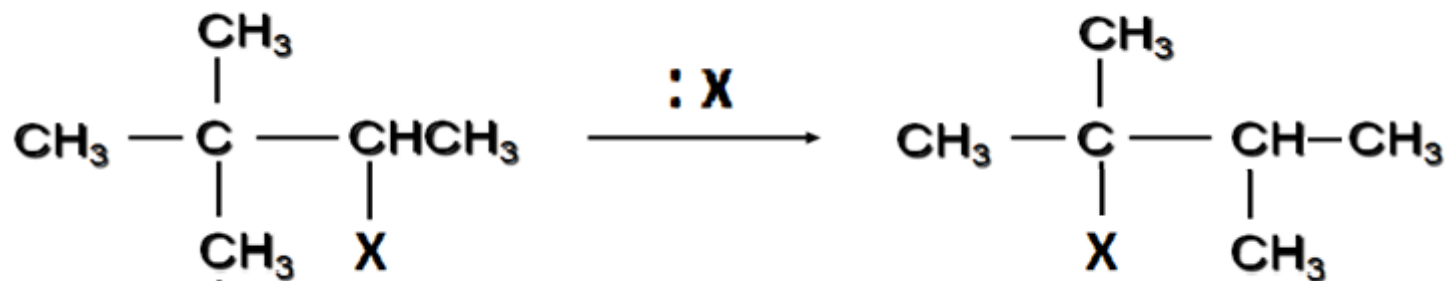
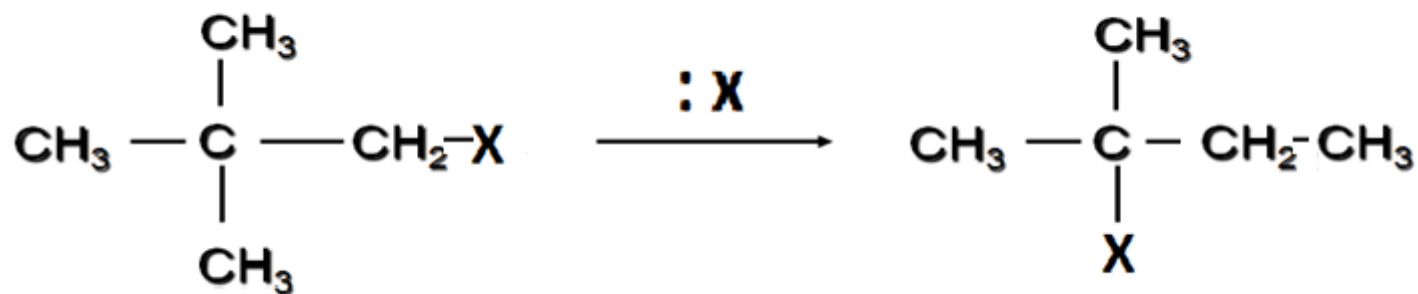


Carbocation Rearrangements in S_N1 Reactions

- It is observed that the entering group attached to a different carbon atom than one that of original position.



Rearrangement in the C skeleton

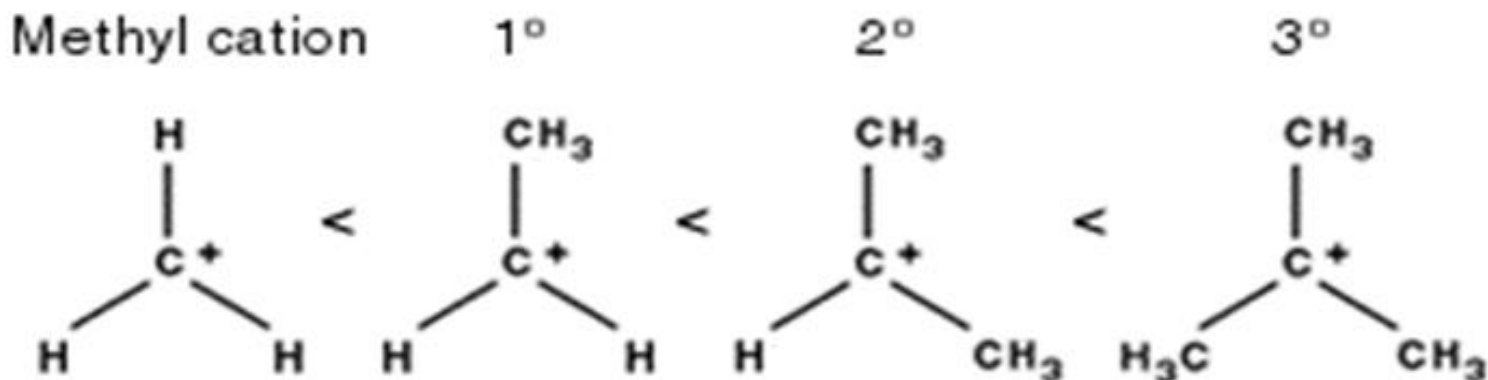


- Carbocations are intermediates in S_N1 reactions, therefore rearrangements are possible
- Less stable carbocations converted to more stable forms by migrating 'H' atoms, alkyl group with an electron pair from an adjacent 'C' atom, to the C having the +ve charge..

Migration of 'H' atom = hydride shift

Migration of 'C' atom = alkyl shift

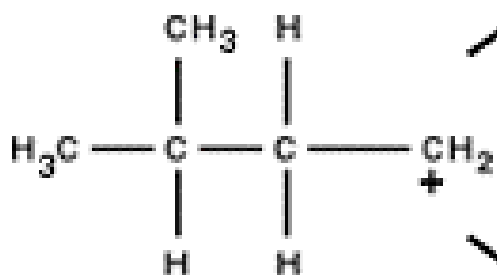
Carbocation stability



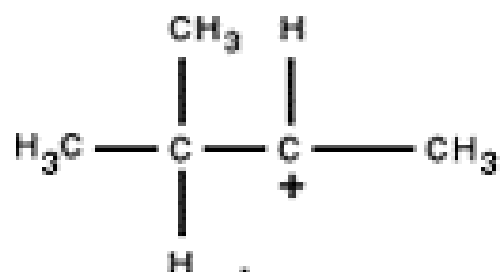
Carbocations rearrange to the more stable form(s)

Carbocation rearrangements

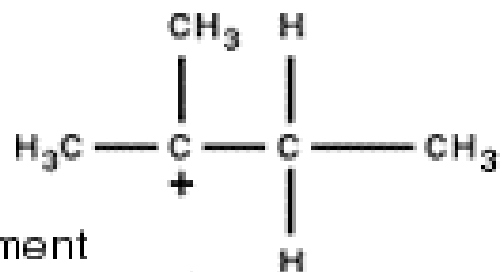
Primary carbocation



Secondary carbocation

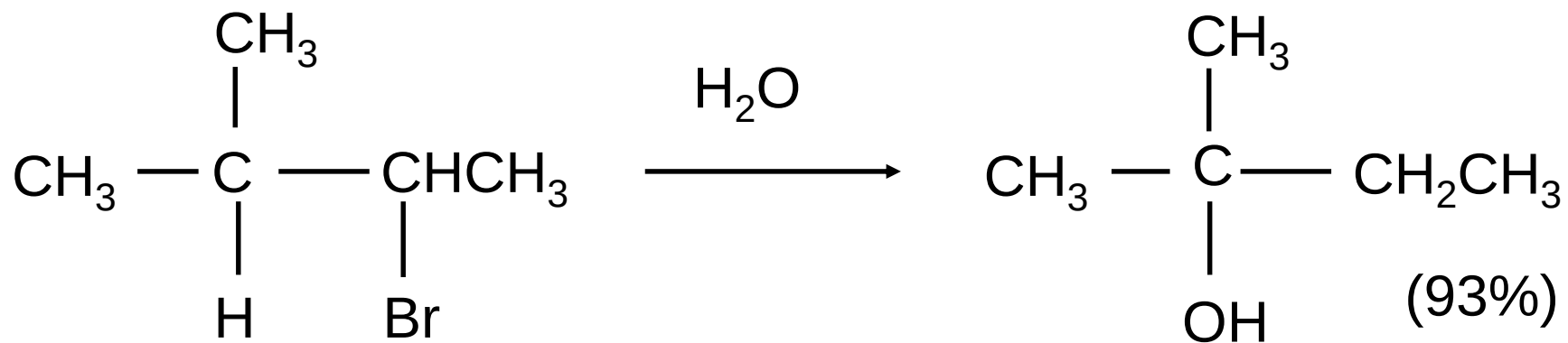


Tertiary carbocation

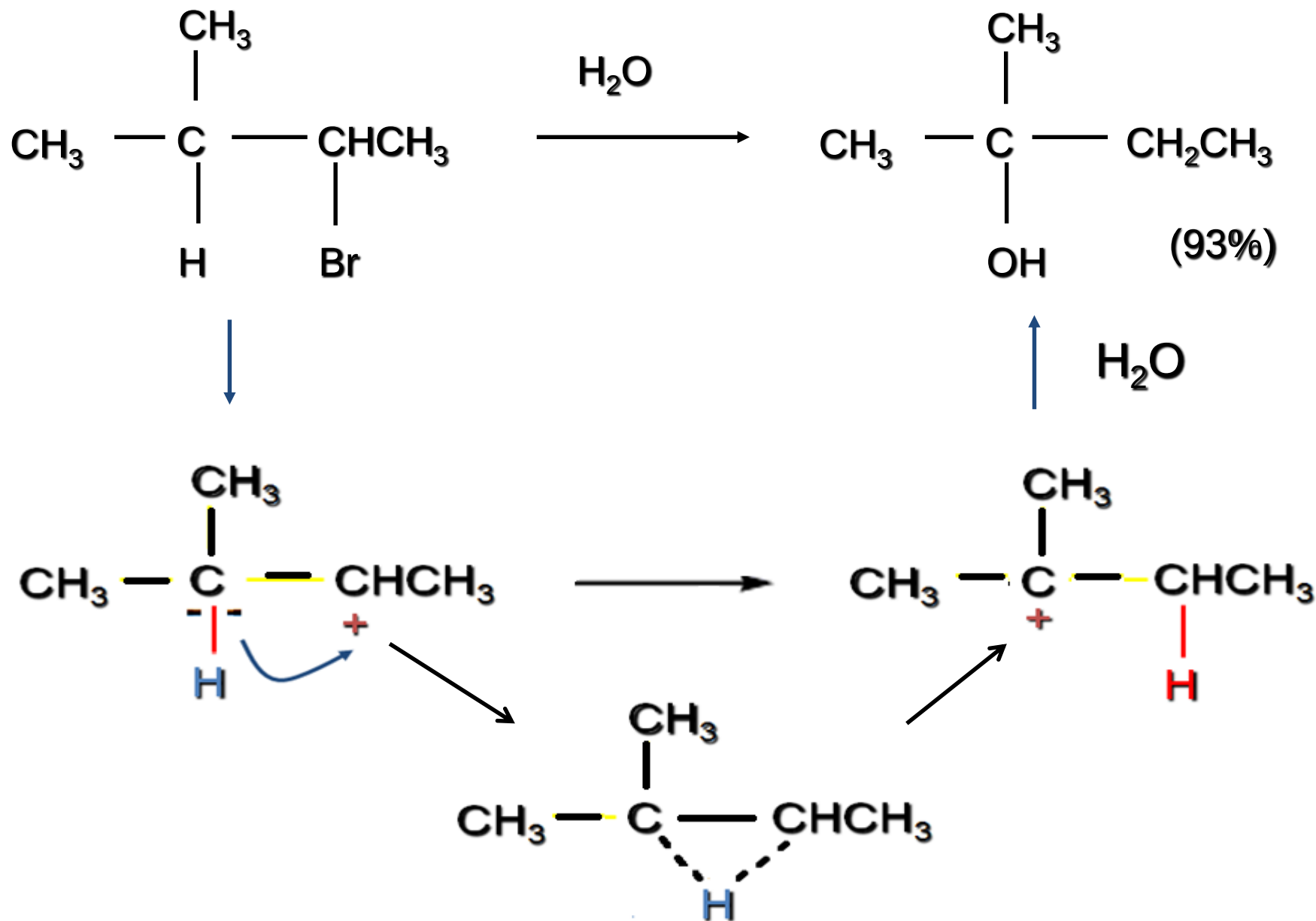


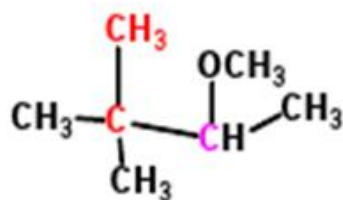
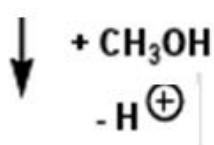
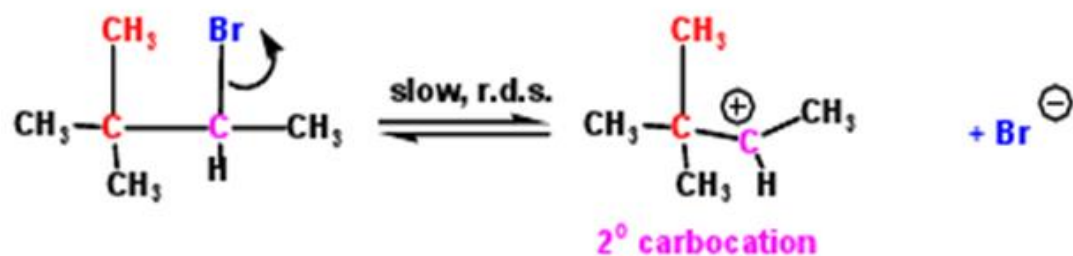
carbocations undergo rearrangement
to form a more stable carbocation

Example

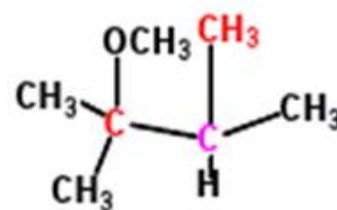
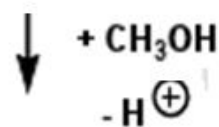


Example





3-methoxy-2,2-dimethyl... yield 4%



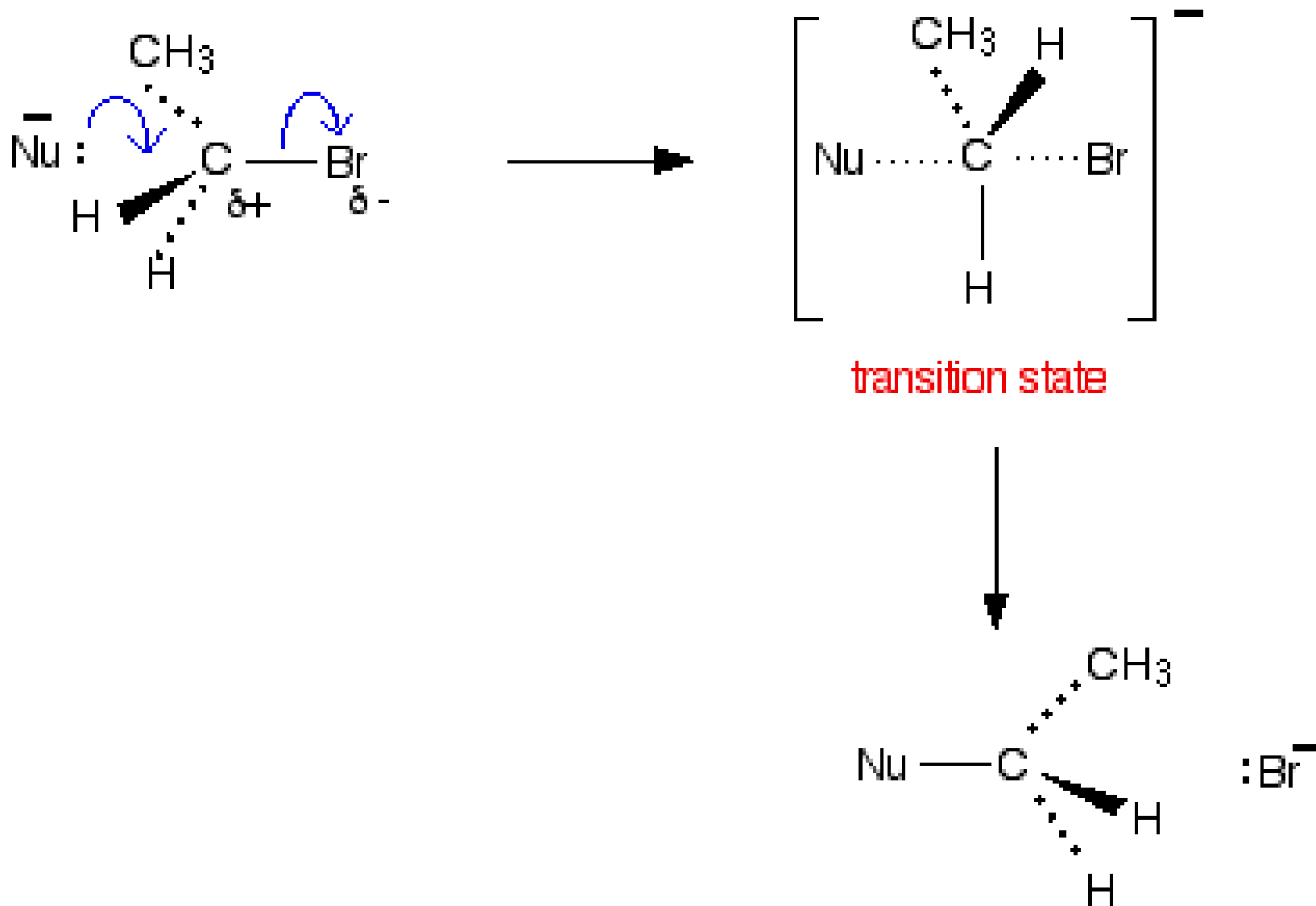
2-methoxy-2,3-dimethyl... yield 41%

Nucleophilic Substitution

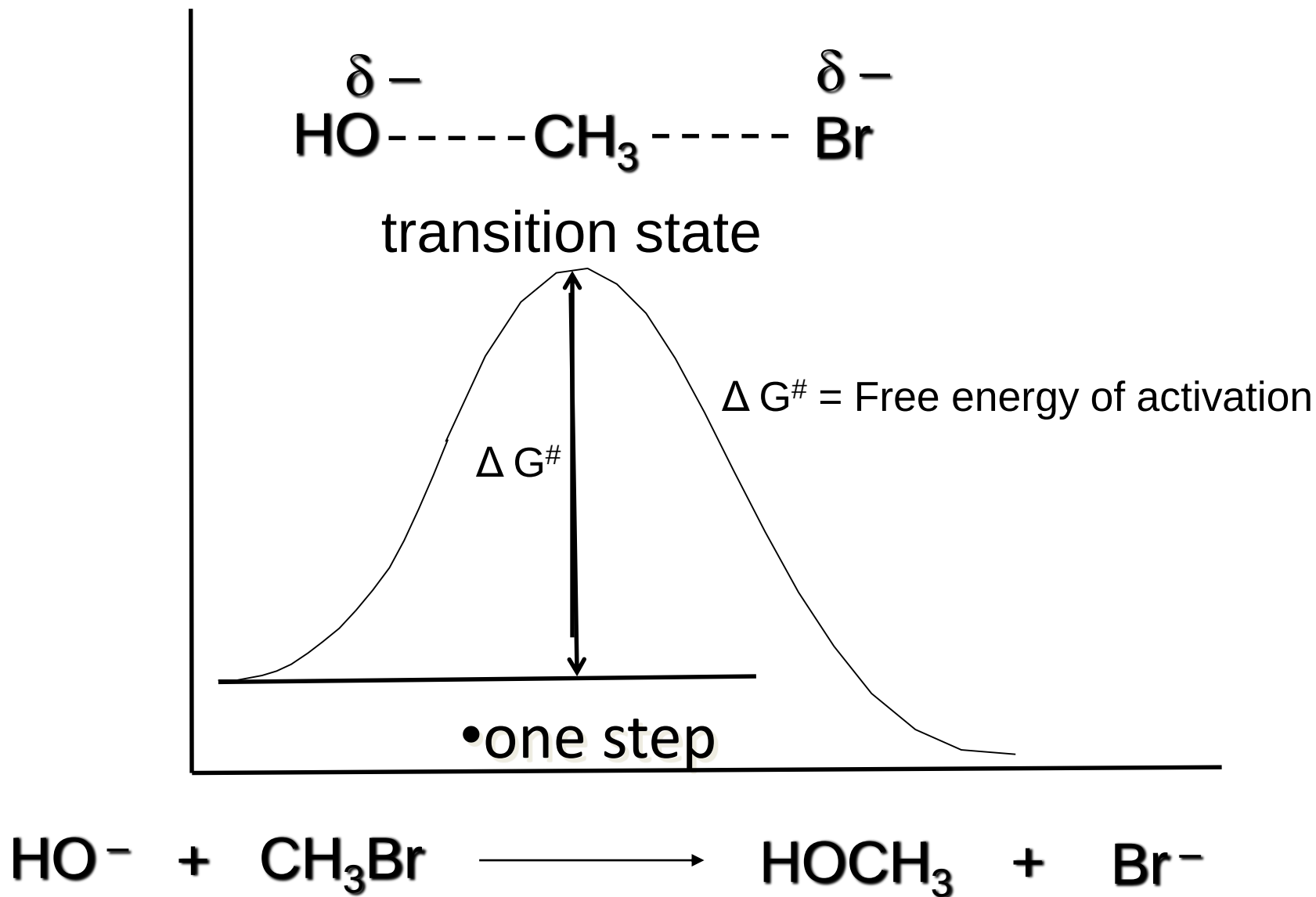
S_N2 Mechanism

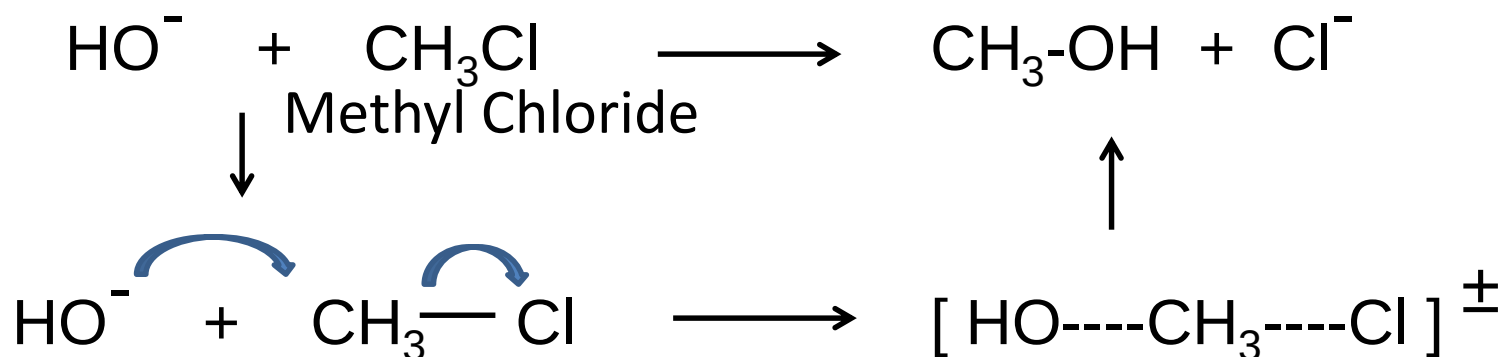
- “S & N” stands for 2 indicates the reaction is bimolecular i.e. the nucleophile and the substrate are involved in the key step.
- It's a one step process.
- The nucleophile attacks from the back side of C – L bond.
- Nucleophile and the leaving group partly bonded to the carbon at transition state (no intermediates).
- Then departure of the leaving group.

S_N2 Mechanism



Bimolecular mechanism





$$\text{Rate} = K_r = [\text{MeCl}] [\text{OH}]$$

The reaction rate depends on the concentration of both molecules.

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Factors influencing Rate of S_N2 Reaction

- Concentration of the nucleophile and concentration of substrate increases the rate

Factors influencing Rate of S_N2 Reaction....

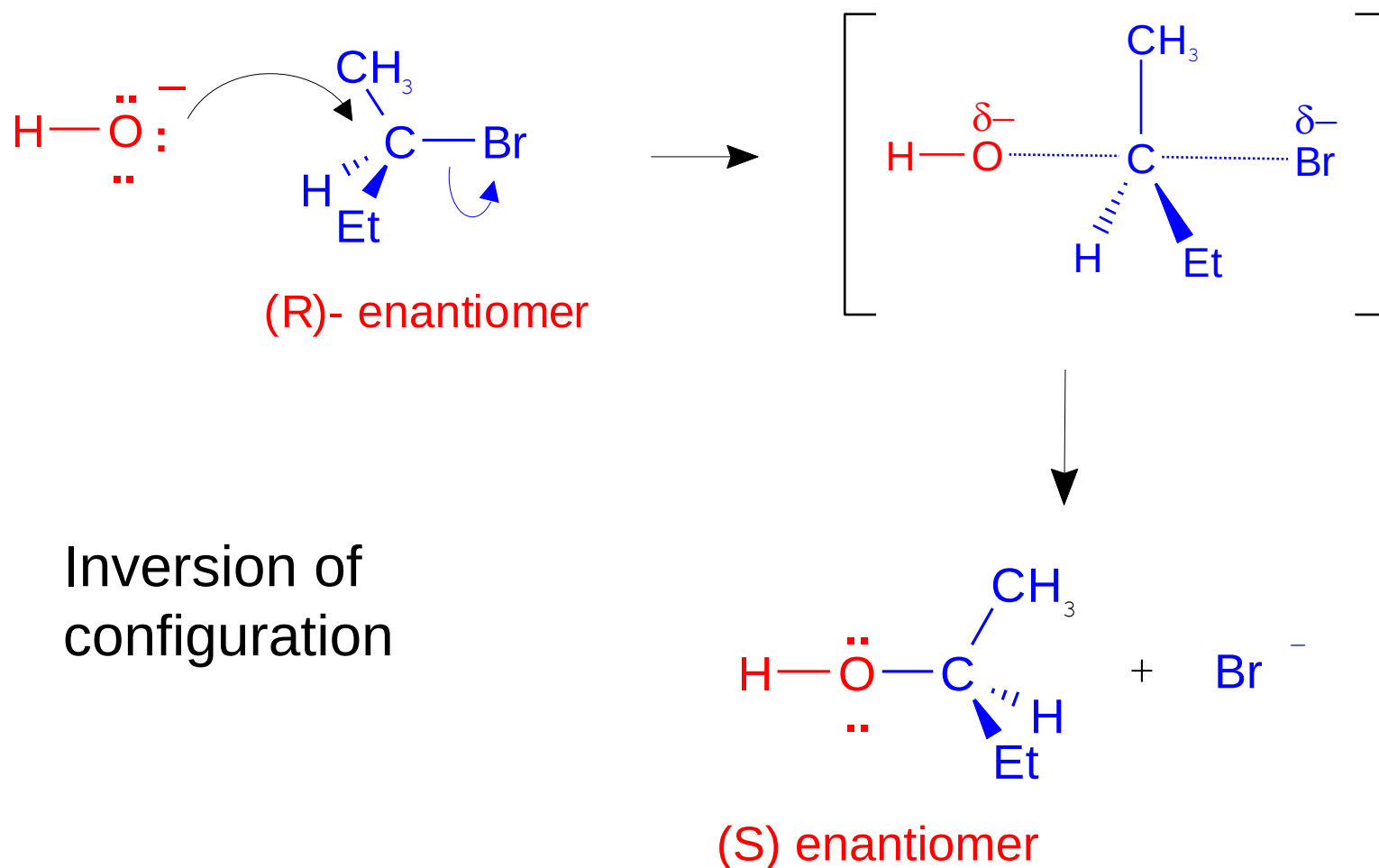
- The bimolecular mechanism proceeds through a highly sterically hindered transition state, where five groups are bonded to the carbon. Therefore the larger groups causes higher strain energy, slower the reaction. Reaction is fastest when the alkyl halide is primary; intermediate for secondary and slower for tertiary.

Because the formation of the transition state, it is easy for primary compare to tertiary.



S_N2 Reaction: stereochemistry

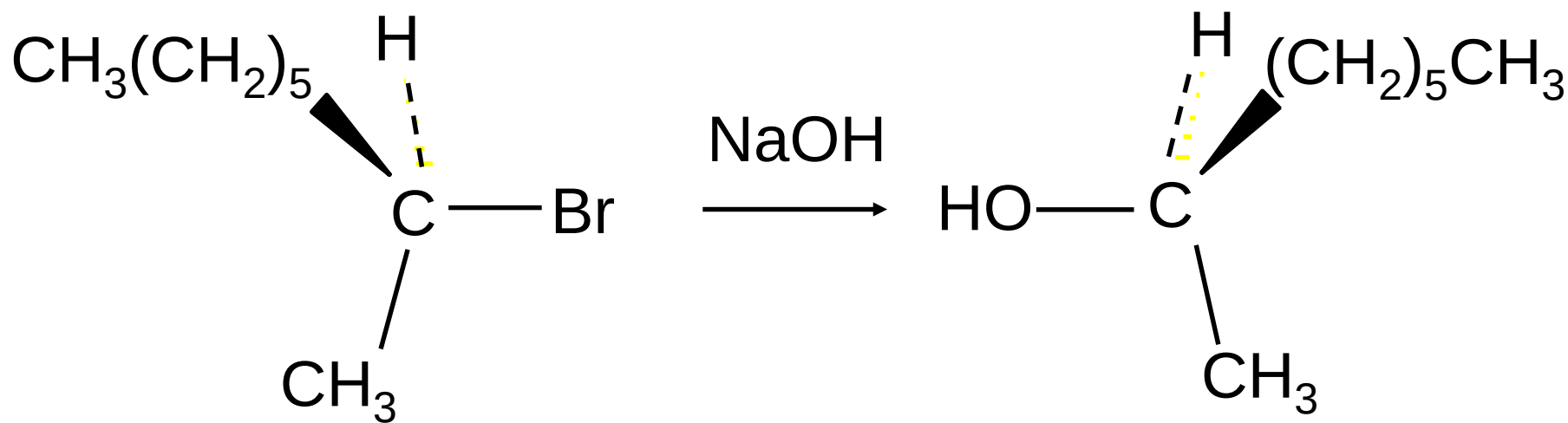
Nucleophilic substitutions S_N2 are stereospecific and proceed with inversion of configuration.



- The reaction of 2-bromooctane with NaOH (in ethanol-water) is stereospecific.

(+)-2-Bromooctane $\rightarrow$ (–)-2-Octanol

(–)-2-Bromooctane $\rightarrow$ (+)-2-Octanol



(*S*)-(+)-2-Bromooctane

(*R*)-(–)-2-Octanol

S_N1 and S_N2 Comparison

- Polar protic solvents favor S_N1 reaction
e.g. H_2O
- Polar aprotic solvent favor S_N2 reaction
e.g. Acetone, DMF, DMSO
- If the nucleophile is strong; then S_N2 is favored.
 1. $HO^- > HOH$
 2. $HS^- > HO^-$
 3. $NH_3 > H_2O > HF$

S_N1 and S_N2 Comparison

- If the carbon bearing the leaving group is stereogenic:
 1. S_N1 reaction occurs with loss of optical activity (that is, with racemization)
 2. S_N2 reaction occurs with inversion of configuration.